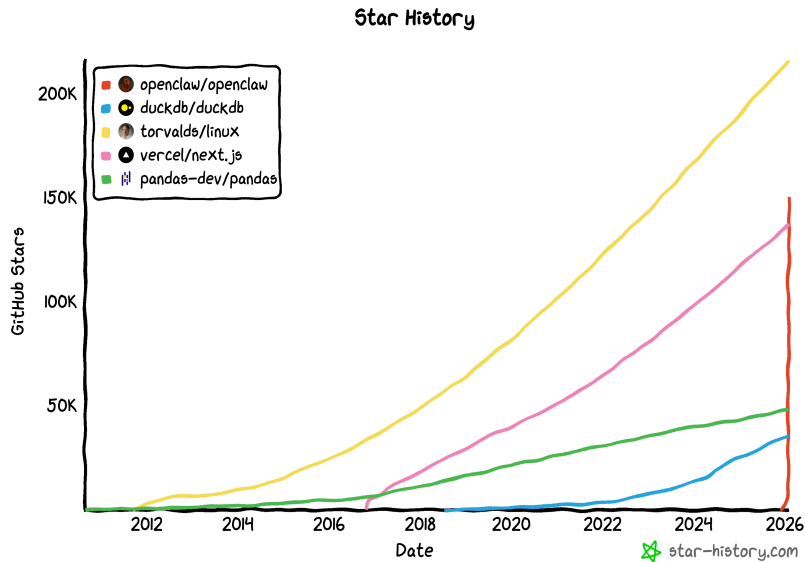


The Directions of Technical Change

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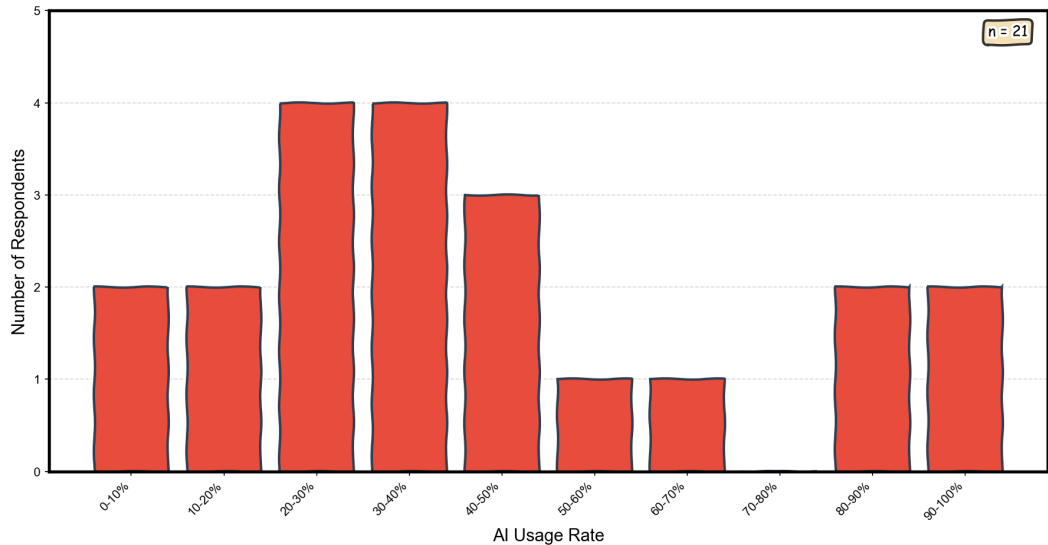
2026

One of these is an AI tool



AI adoption at CEU Economics differs widely

How much of your core working time do you spend working with AI tools?



How can both be true?

If AI generates near-instant adoption...

... why doesn't everybody adopt it?

Mixed evidence on productivity, wage, and employment effects of AI (Brynjolfsson et al. 2025, Humlum & Vestergaard 2025, Bick et al. 2024, Becker et al. 2025).

A model of digital technology

1. Technology is **directional** ($t \in \mathbb{R}^N$)
2. Adoption is **endogenous** (someone is maximizing something)
3. Workers have some **flexibility** in doing their job

Why existing models are insufficient

Too few dimensions

skilled/unskilled, complex/routine, expert > novice

Too rigid

Technology capabilities taken as given. Workers cannot change what they do.

The model

The worker's problem

$$\max_x F(x) = \left(\sum_{i=1}^N \theta_i^{1/\sigma} x_i^{1-1/\sigma} \right)^{\sigma/(\sigma-1)}$$

subject to

$$\left(\sum_{i=1}^N s_i^{-1/\gamma} x_i^{1+1/\gamma} \right)^{\gamma/(\gamma+1)} \leq B$$

x = activities, s = skills, θ = job requirements, B = budget (time in a day).

Solution

Activity levels:

$$x_i^A = B \frac{s_i^{\gamma/(\gamma+\sigma)} \theta_i^{\sigma/(\gamma+\sigma)}}{\left[\sum_j s_j^{(\sigma-1)/(\gamma+\sigma)} \theta_j^{(\gamma+1)/(\gamma+\sigma)} \right]^{\sigma/(\gamma+\sigma)}}$$

Weighted average of skills and job requirements.

Productivity

$$\frac{Y^A}{B} = \frac{F(x^A)}{B} = \left[\sum_j s_j^{(\sigma-1)/(\gamma+\sigma)} \theta_j^{\gamma+1/(\gamma+\sigma)} \right]^{\sigma/(\sigma-1)}$$

Output per unit budget. Increases in all s_i and all θ_i . Highest when $s_i \propto \theta_i$.

Two parameters capture job flexibility

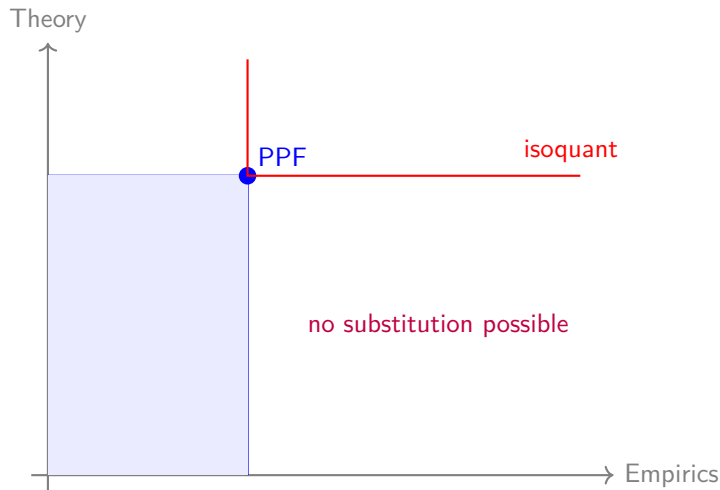
γ = elasticity of transformation (PPF curvature)

σ = elasticity of substitution (isoquant curvature)

Flexibility

- ▶ Knowledge workers choose how to allocate time
- ▶ “Shadow IT”: workers use preferred tools regardless of policy
- ▶ This model nests the standard case: Leontief → employer dictates

Rigid jobs: no flexibility



Worker produces at the kink. Technology offers empirics, but the job needs **both** in fixed proportions.

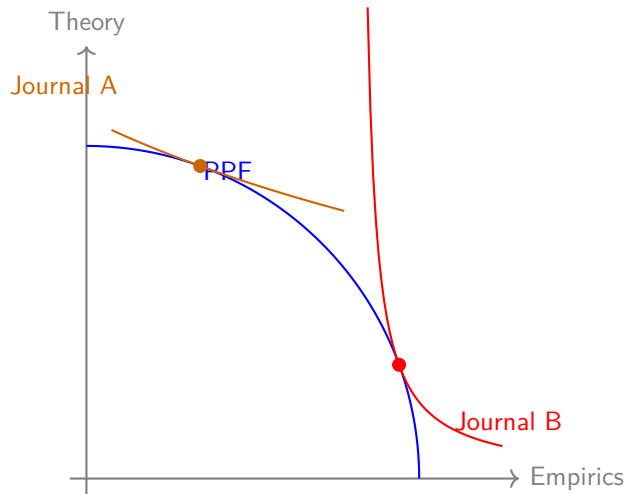
Jevons Paradox

Time share on activity i :

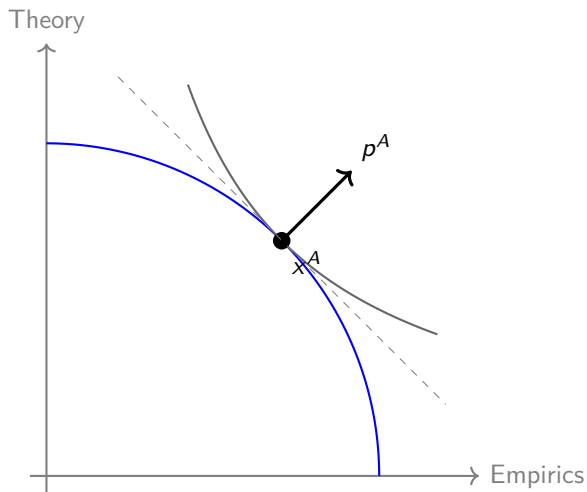
$$\omega_i^A = \frac{s_i^{(\sigma-1)/(\gamma+\sigma)} \theta_i^{(\gamma+1)/(\gamma+\sigma)}}{\sum_j s_j^{(\sigma-1)/(\gamma+\sigma)} \theta_j^{(\gamma+1)/(\gamma+\sigma)}}$$

Increases in skill s_i if $\sigma > 1$. Relevant range for knowledge work.

Different journals have different requirements



Autarky prices



At the optimum, the PPF and isoquant are tangent.

The supporting hyperplane defines **autarky prices** p^A .

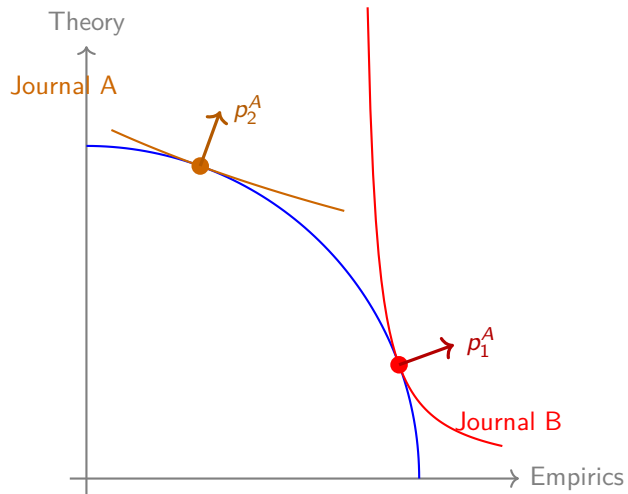
Autarky prices (cont'd)

$$p_i^A \propto \left(\frac{\theta_i}{s_i} \right)^{1/(\gamma+\sigma)}$$

High for activities that are required (θ_i high) but hard to do (s_i low).

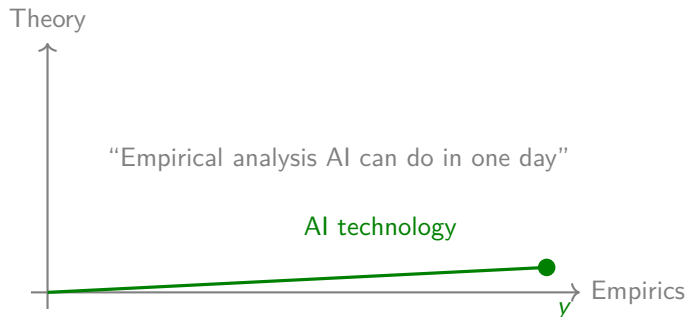
Inference problem: worker does an activity x_i either because of θ_i or because of s_i .

Autarky prices characterize job activities



Technology

Technology as a spike



Same resource constraint: one day.

Technology enhances activities in a *fixed combination* (directional).

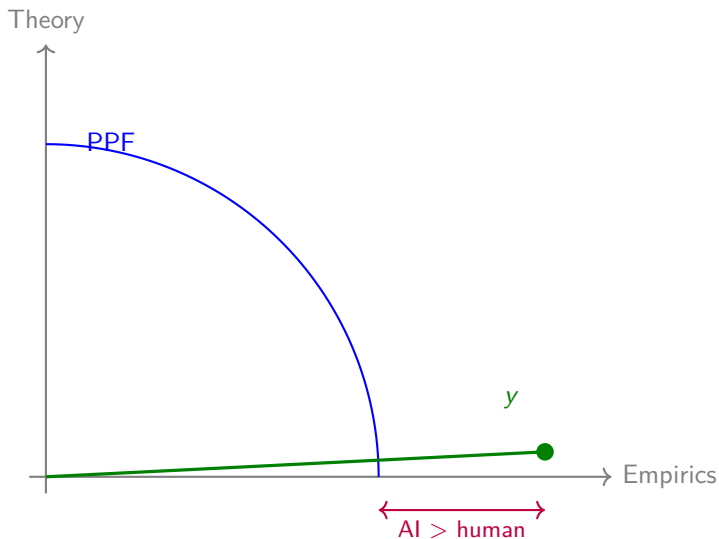
You never use technology alone

The spike is too thin. No paper requires theory and empirics in exactly the t proportion.

You use technology **in combination** with your own skills.

This will remain true with $K > 1$ technologies as long as $K < N$. AI has a **jagged** capability profile: Hendrycks et al. (2025) define 46 intelligence tasks, AI is capable in a small subset.

Technology meets the individual



AI has **absolute advantage** in 90% empirics, 10% theory.

AI + human = convex hull

Using technology takes time (giving tasks, supervising, reviewing results).

Spend λ of your day on AI, $1 - \lambda$ on your own work.

$$\max_{x, \lambda} F((1 - \lambda)x + \lambda y) \quad \text{s.t.} \quad x \in \text{PPS}, 0 \leq \lambda \leq 1$$

Convex hull of the two production possibility sets.

Assumptions: technology adoption

Time use of tech and human are perfect substitutes

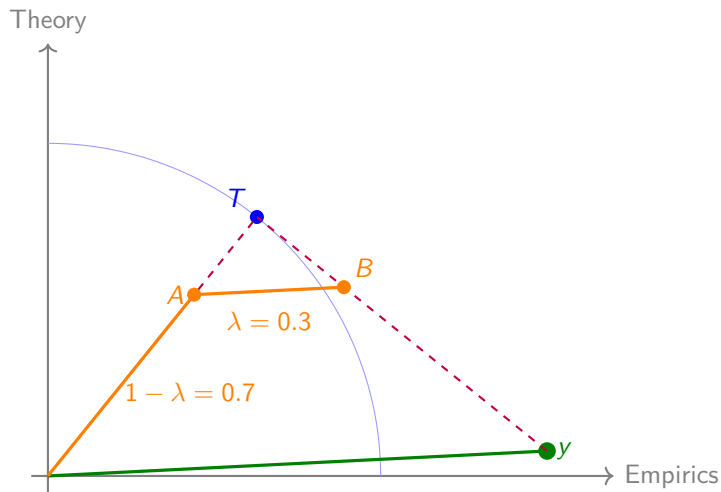
Goodman et al (2026): elastic substitution of time across different online activities.
Consistent with rapid adoption of “vibe coding” (Koren, Békés, Hinz, Lohmann, 2026).

No skill differences in using AI

There are skill differences in **benefiting from** AI.

Everyone can open ChatGPT for free. The question is **leverage**.

AI + human = convex hull



$\lambda \in [0, 1]$ is a choice variable. May be 0 or 1.

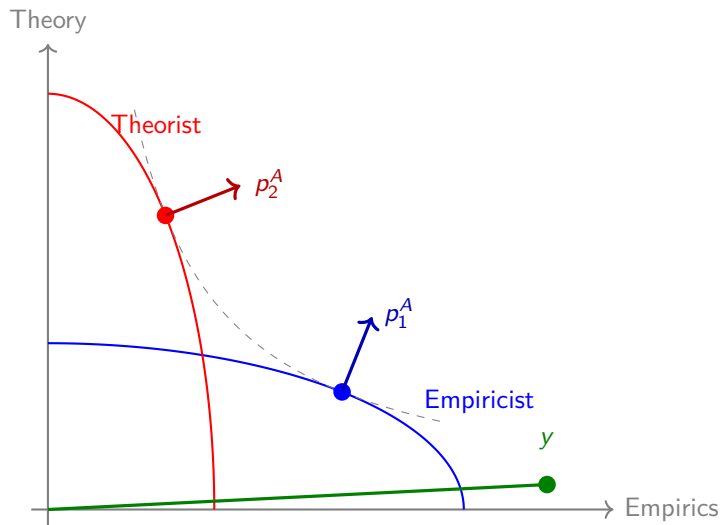
Three scales of technical change

Three scales

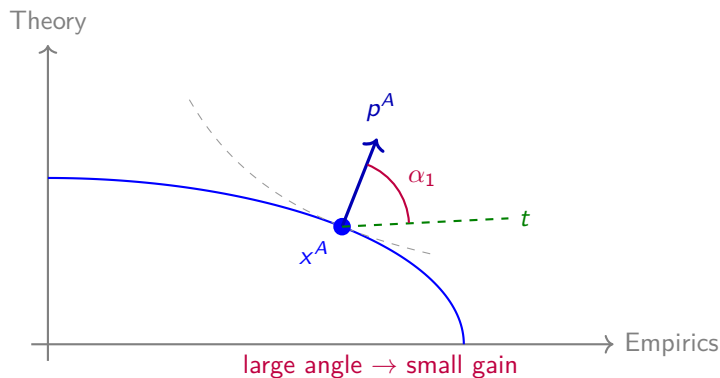
1. **Small-scale:** marginal $\rightarrow$ only income effect
2. **Medium-scale:** changes activities $\rightarrow$ price effect
3. **Large-scale:** changes sorting across jobs (not today)

Small-scale technology

Two economists



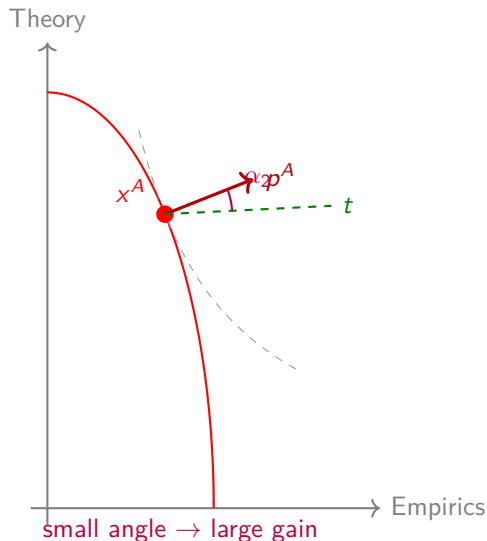
Worker 1: the empiricist



The empiricist's autarky price points toward theory (the expensive activity).

Technology is nearly parallel to what is already **cheap**.

Worker 2: the theorist



The theorist's autarky price points toward empirics (the expensive activity).

Technology is nearly parallel to what is **expensive**: big gain.

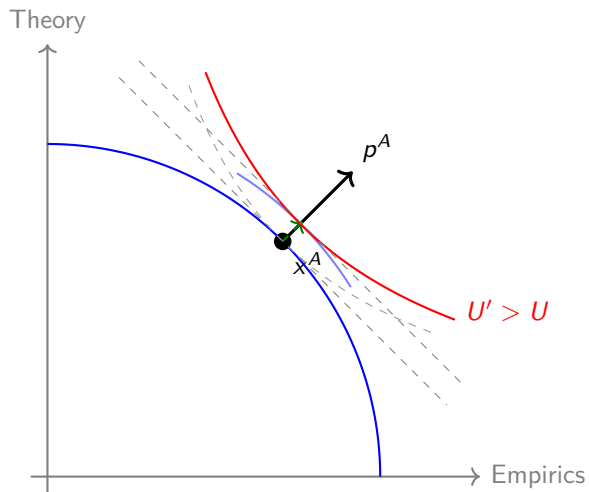
The angle determines everything

For small technology adoption:

$$\text{Productivity gain} \propto p^A \cdot t = \|p^A\| \|t\| \cos \alpha$$

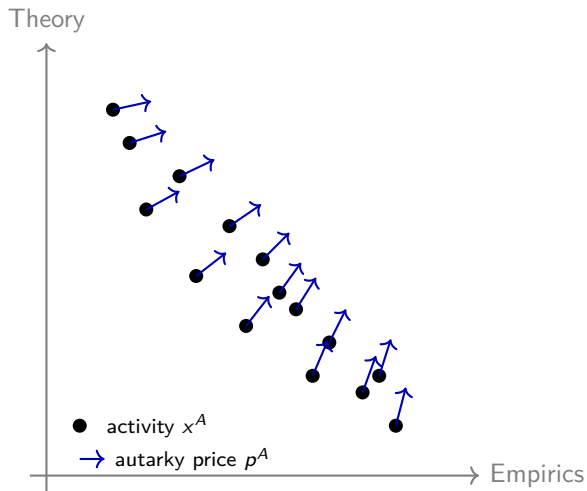
- ▶ Small $\alpha \rightarrow$ large gain (technology targets your weakness)
- ▶ Large $\alpha \rightarrow$ small gain (technology targets your strength)

Best technology direction



The optimal technology direction is p^A .

A population of workers



Each worker's p^A points toward their **expensive** activities.

Assumptions: small-scale technology

A benevolent social planner introduces technology:

1. **Free:** no budget cost (supervising is $o(\varepsilon)$)
2. **Small:** marginal expansion of PPS in one direction
3. **No reallocation:** workers don't change what they do

First-best technology directions

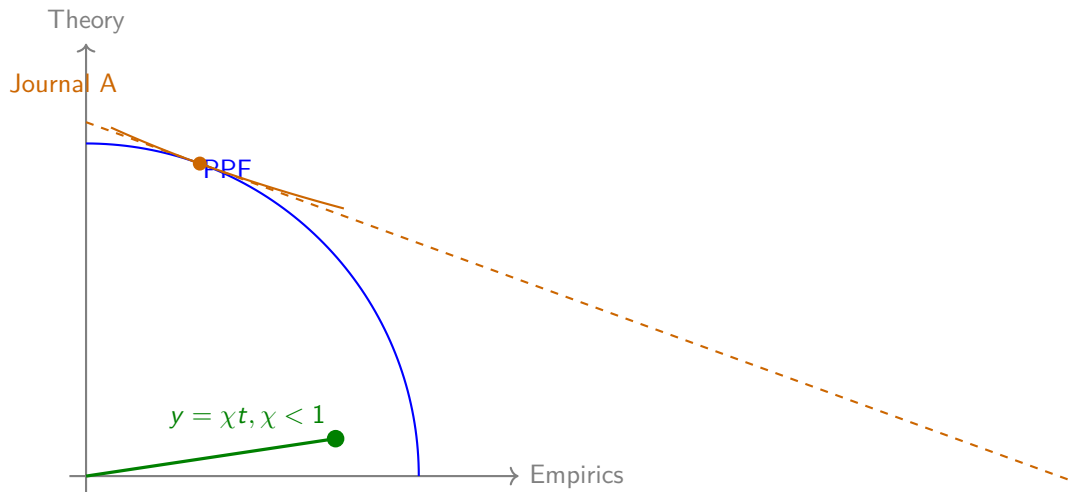
Welfare theorem

Let $M_{p^A} = \mathbb{E}[B \cdot p^A p^{A'}]$, the *second-moment matrix* of autarky prices weighted by budget.

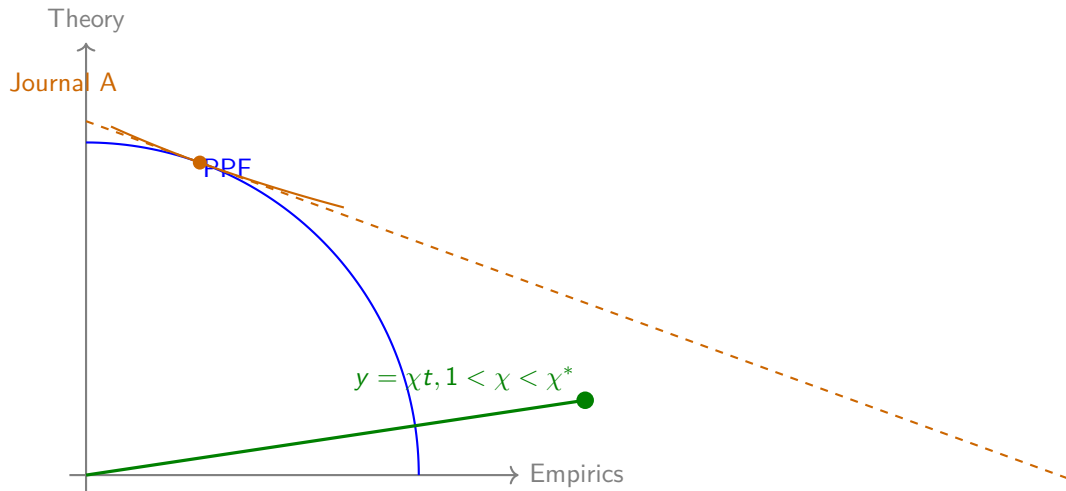
The first-best K technology directions are the K **largest eigenvectors** (“principal components”) of M_{p^A} .

Medium-scale technology

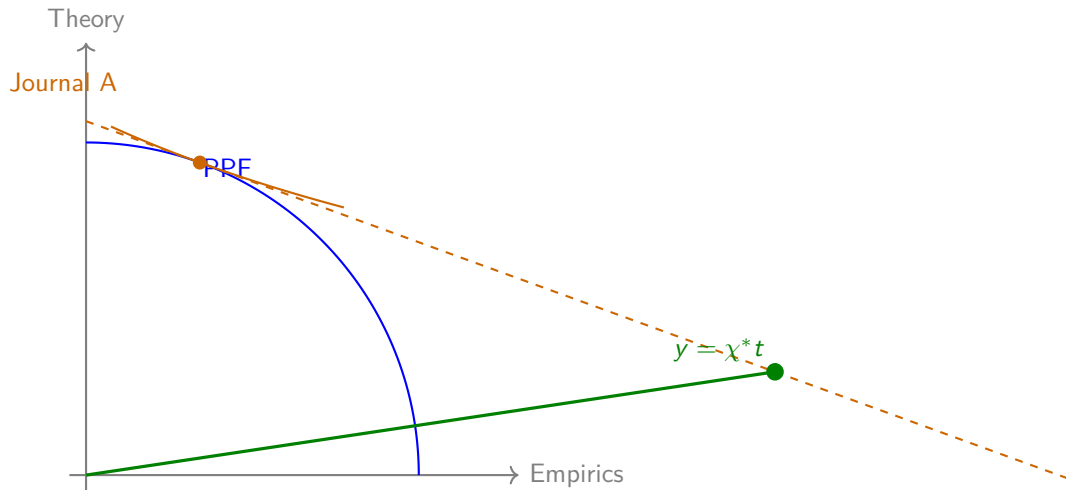
AI has no absolute advantage in anything



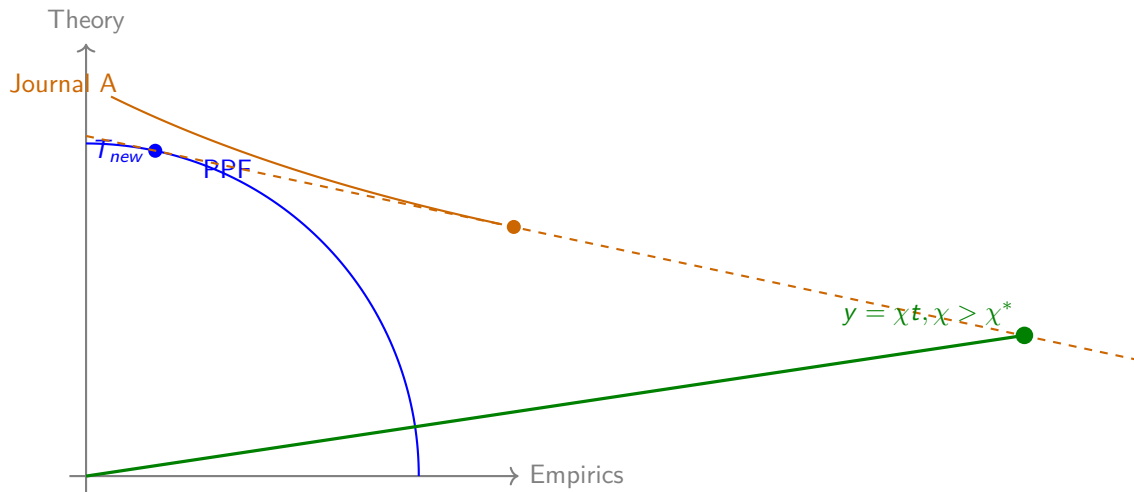
AI has an absolute, but not comparative advantage



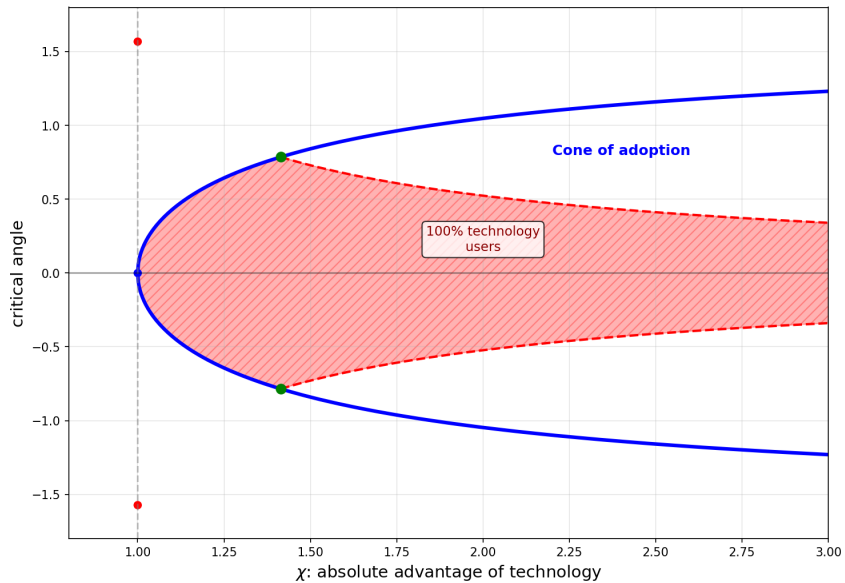
Worker starts adopting AI



Worker specializes in their comparative advantage



The cone of technology adoption



As technology capability increases

1. Threshold effect: Absolute advantage in *some* human activity is necessary but not sufficient for adoption
2. Rapid adoption after the threshold
3. Early adopters go all in: use technology for everything
4. Late adopters remain on the fringe: use technology less intensively
 - ▶ Work activities become *less similar* across workers who adopt
 - ▶ Output become *more similar*
5. Early adopters reduce their intensity of use once technology is widespread.
6. Some people never adopt ($N > K$)

Economic intuition

Income effect from technology

Absolute advantage increases productivity, but only for aligned workers.

Substitution effect from technology

Less aligned workers see a change in their *shadow prices*, leaning in to their **comparative advantage**.

Dynamics

Income effect is immediate → rapid adoption after threshold. Substitution effect depends on alignment → heterogeneous intensity of adoption. Less intensive use once technology is widespread.

Welfare and inequality

1. Who **sets the direction** of technology?
2. Who **sets the scale** of technology?
3. Who **decides to adopt**?
4. Who **reaps the benefit**?

Today, we have taken (1) and (2) as given and assumed frictionless on-the-job decisions/negotiations for (3) and (4).

Applications

Theoretical applications

1. Directed technical change
2. Sorting across occupations, GE effects
3. Collaboration
4. Marketable activities

Empirical applications

1. Measure θ and s from O*NET or similar
2. Estimate γ and σ from how workers change what they do
 - ▶ How are jobs reshaped from within?
 - ▶ Which jobs will disappear with the next wave of AI?
3. Focus on a particular industry (e.g. software engineering)

Summary